Maulana Azad National Institute of Technology, Bhopal

Department of Mathematics, Bioinformatics and Computer Application
Session 2024 - 2025



Low Light Image Enhancement

Minor Project (CA – 406)
MCA IV Sem

Members Name:
Vishal Jatav _ 2320403236
Bibek Nayak _ 2320403250

Project Mentor:
Prof. Amit Bhagat Sir

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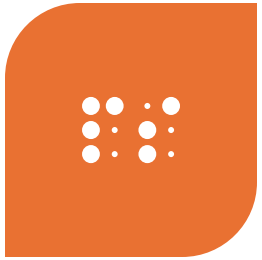
Session 2024 - 2025

Low Light Image Enhancement

- A Comparative Study of Enhancement Techniques



Introduction to Low-Light Image Enhancement



LOW-LIGHT CONDITIONS IN IMAGING REFER TO ENVIRONMENTS WITH INSUFFICIENT ILLUMINATION, CAUSING POOR VISIBILITY, HIGH NOISE, LOW CONTRAST, AND LOSS OF DETAILS IN CAPTURED IMAGES



CHALLENGES POSED BY LOW-LIGHT IMAGES (LOW BRIGHTNESS, CONTRAST LOSS, NOISE)



APPLICATIONS: SURVEILLANCE, MEDICAL IMAGING, REMOTE SENSING, NIGHT PHOTOGRAPHY

Classification of Low-Light Enhancement Methods



Seven key categories:



Gray transformation methods



Histogram equalization (HE) methods



Retinex-based methods



Frequency-domain methods



Image fusion methods



Defogging-based methods



Machine learning-based methods

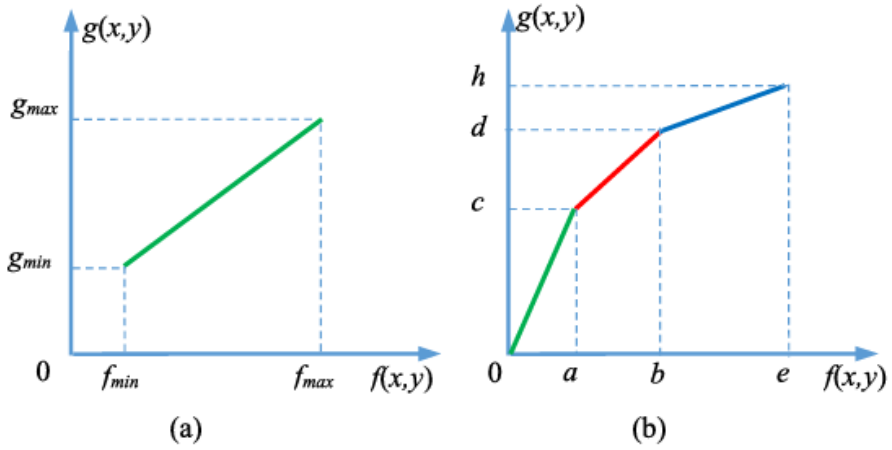
Gray Transformation Methods

Overview: Pixel-based brightness adjustments

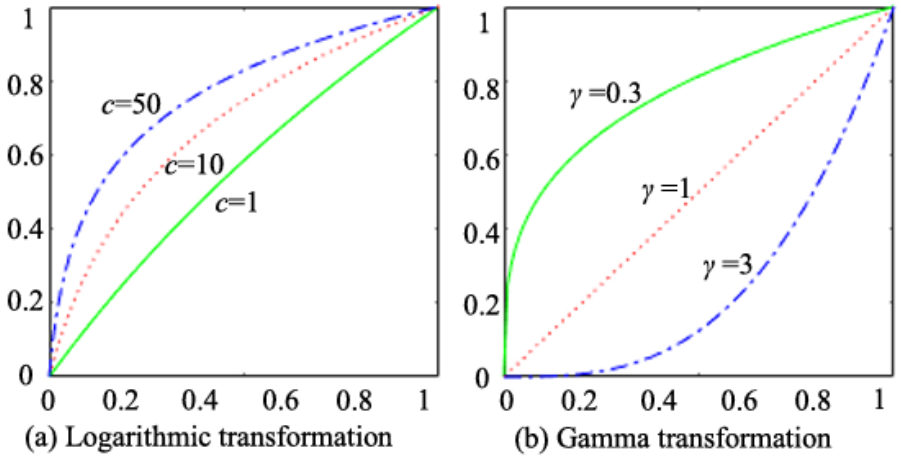
Techniques: Linear vs. Nonlinear transformations

Strengths: Fast processing, simple

Limitations: May cause over-enhancement, unnatural results



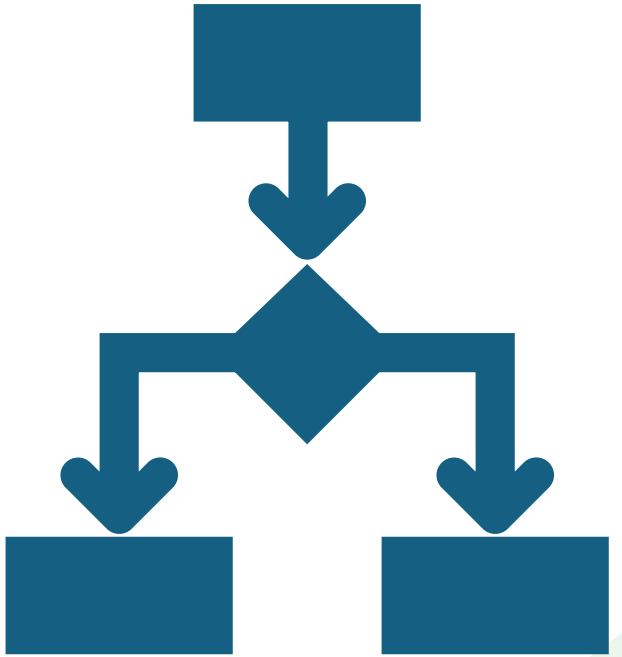
Linear transformation curves

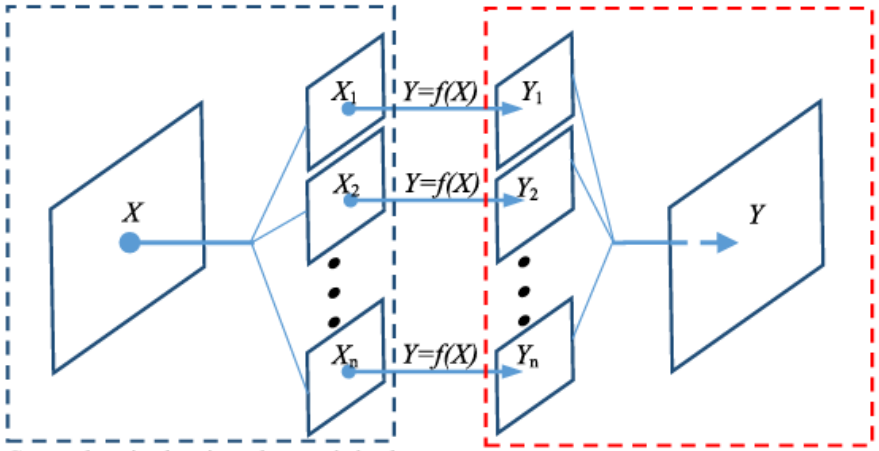


Non-Linear transformation curves

Histogram Equalization (HE) Methods

- Concept: Spreads intensity values for better contrast
- Variants:
  - Global HE (GHE)
  - Local HE (LHE)
  - Adaptive HE (CLAHE)
- Limitations: Color distortion, noise amplification





Basic model of GHE algorithms

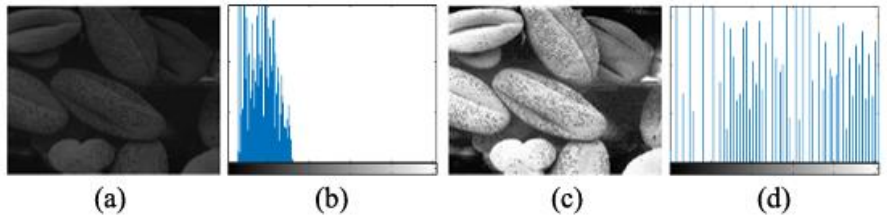
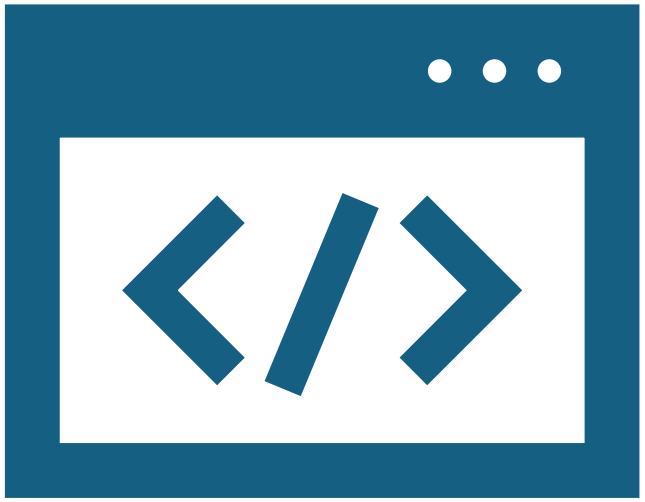


FIGURE 7. **Example of HE on a grayscale image.**

Example of HE on grayscale image



Retinex-Based Methods

- Inspired by human visual perception
- Enhances details while preserving natural color

Algorithms:

- Single-Scale Retinex (SSR)
- Multi-Scale Retinex (MSR)
- MSR with Color Restoration (MSRCR)

$$I(x,y) = R(x,y)\,L(x,y)$$

- L(x, y) is the illumination component, which depends on the environmental light characteristics; and I(x, y) is the received image. L(x, y) determines the dynamic range of the image, whereas R(x, y) determines the inherent nature of the image.

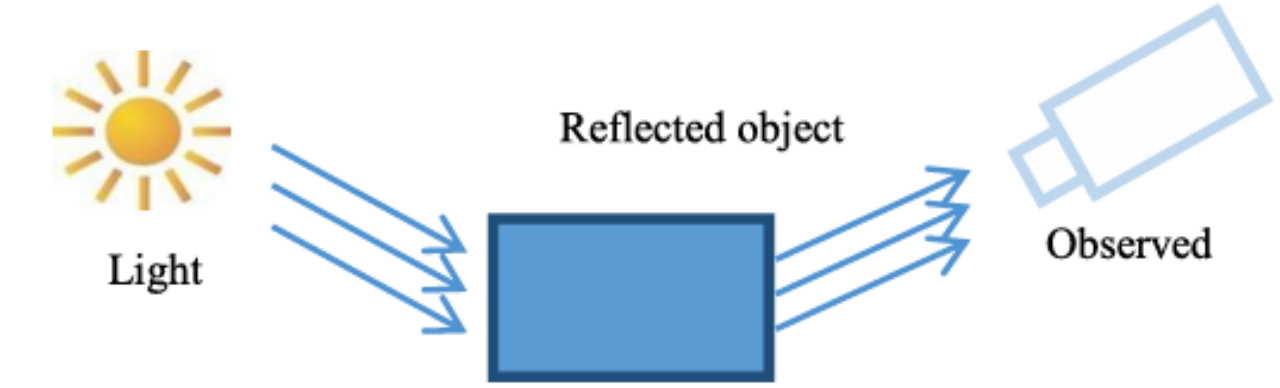
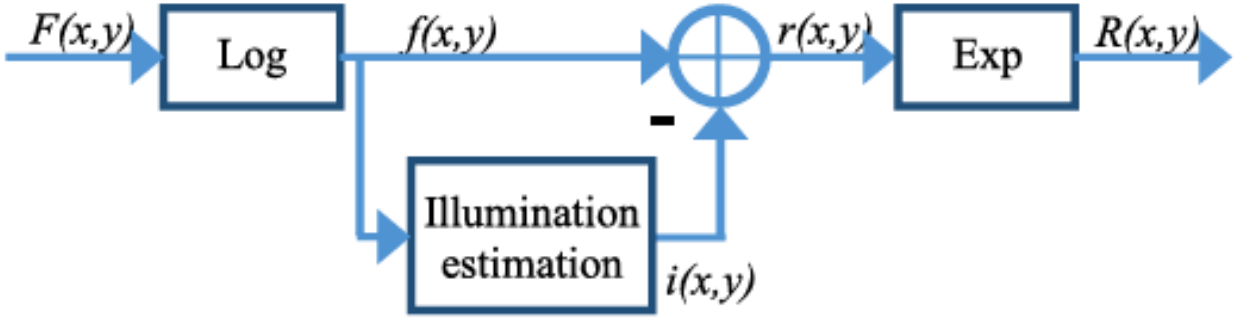


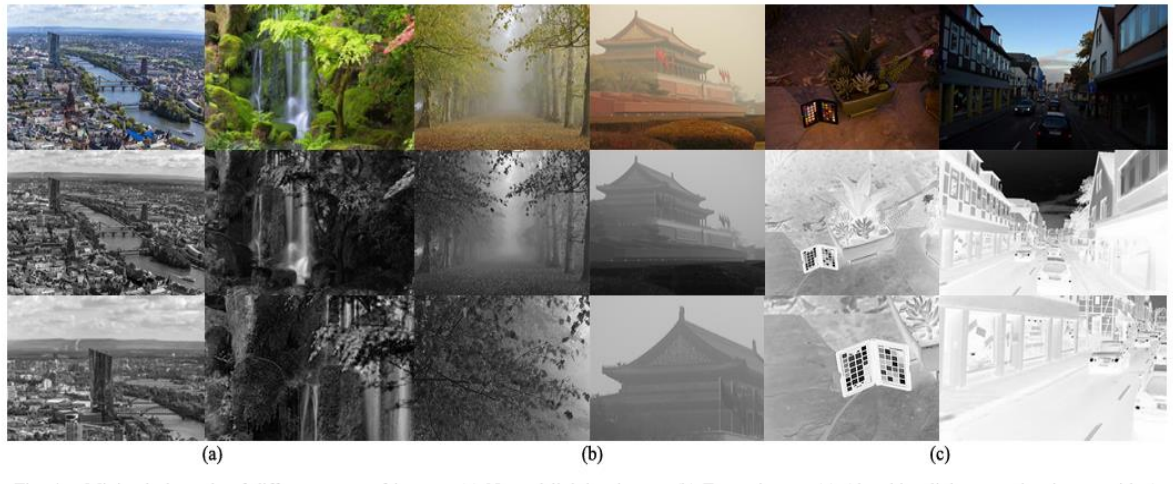
FIGURE 13. Light reflection model.



General process of the Retinex algorithm

The Absorption Light Scattering Model (ALSM)

- Introduces a physics-based model for low-light imaging
- Simulates light absorption and scattering
- Uses atmospheric light and transmittance for enhancement
- Prevents noise amplification at edges

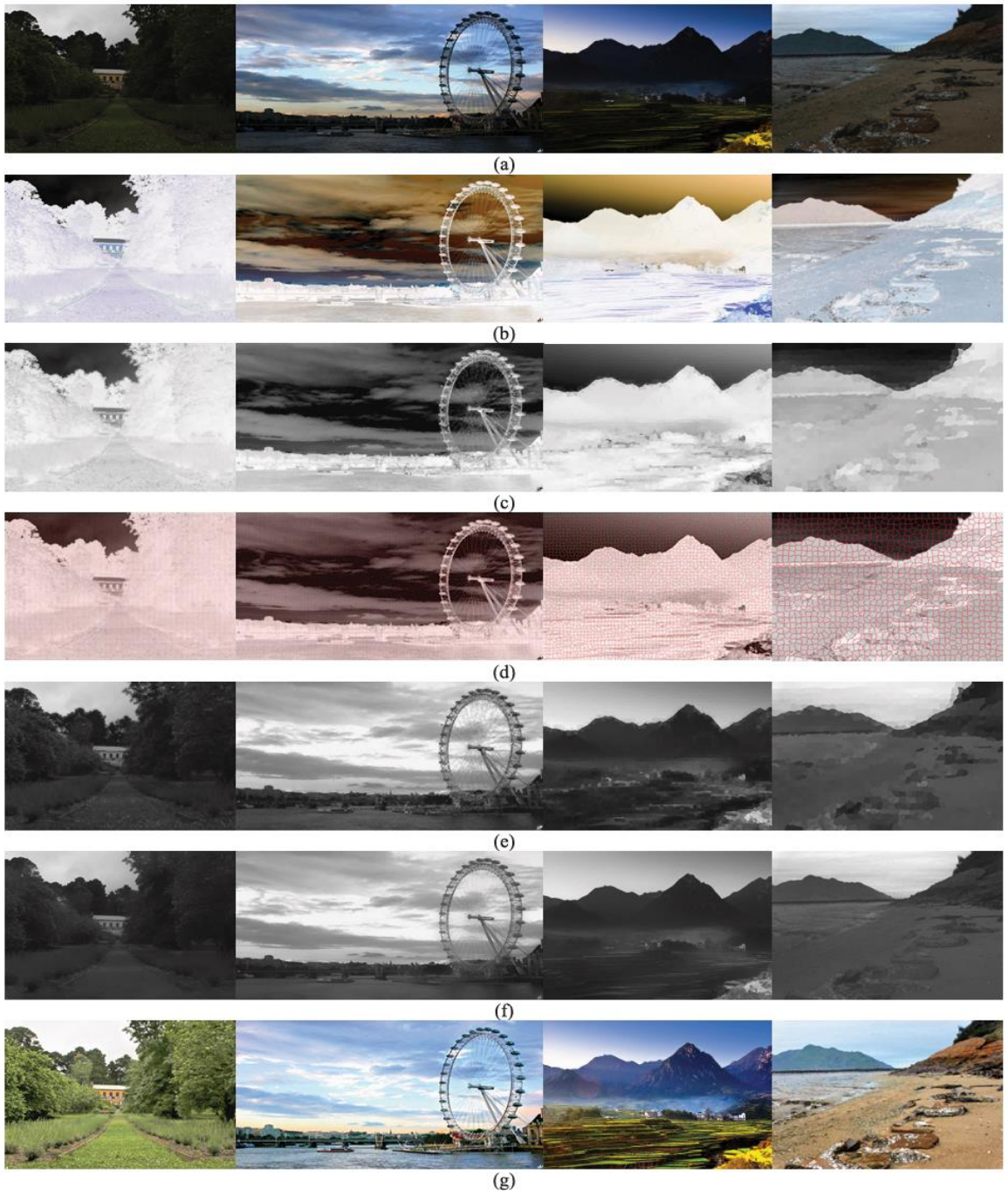


ALSM – Key Components

Minimum channel constraint: Prevents gradient-based noise amplification

Improvements over traditional methods:
- More natural enhancement
- Less color distortion
- Reduced noise

Mean Standard Deviation (MSD) mechanism: Adaptive transmittance estimation



Performance Evaluation



Quality metrics for enhanced images:

Contrast improvement

Noise suppression

Color fidelity



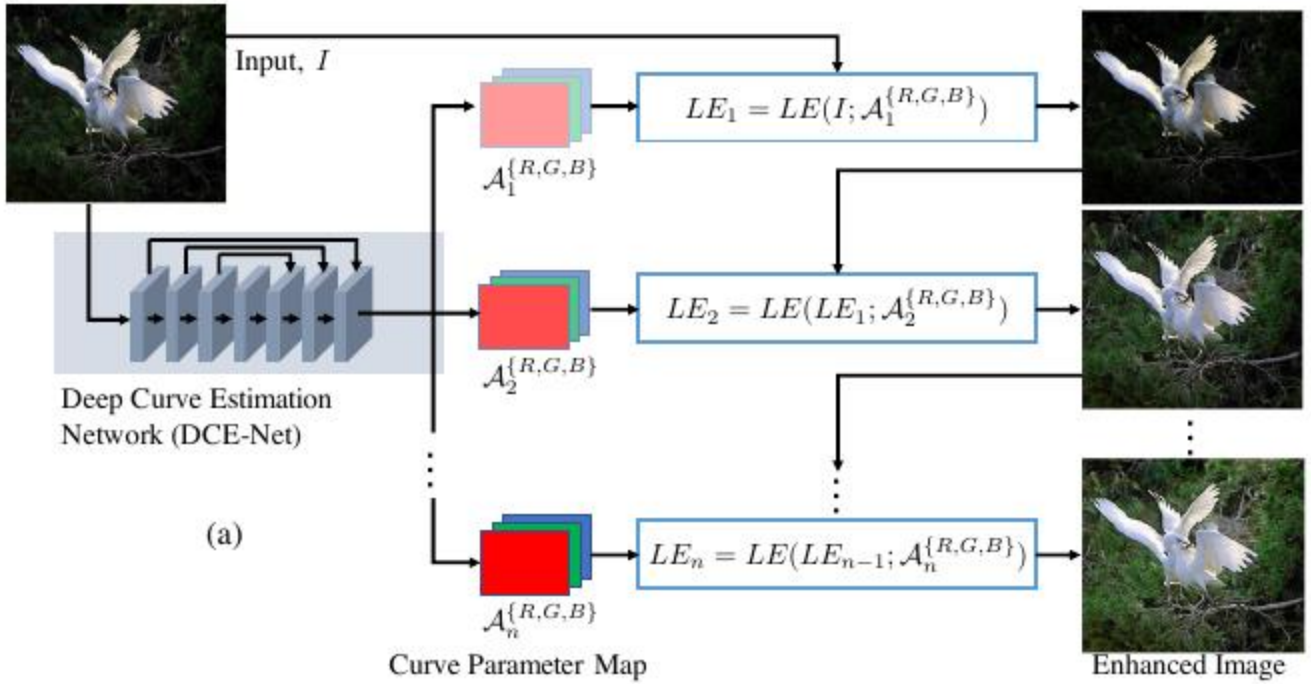
ALSM vs. other methods: Better balance of enhancement and noise control

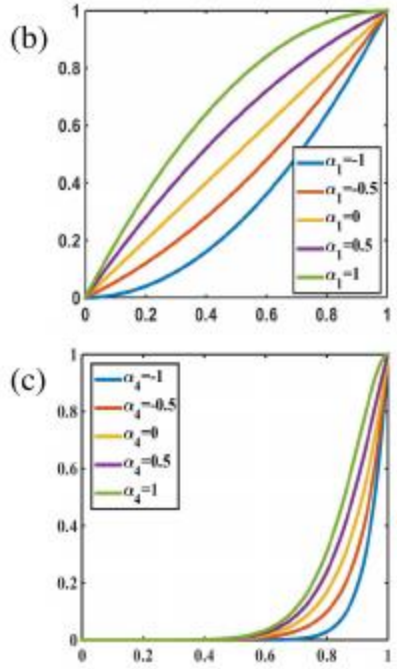
The Zero-DCE Framework

Goal: Estimate optimal light-enhancement curves (LE-curves) for image enhancement.

Process: Iteratively map RGB pixels using LE-curves to enhance low-light images.

Understanding Light-Enhancement Curves





Definition: Maps low-light images to enhanced versions using self-adaptive parameters.

Key Objectives:

Maintain pixel values in **[0,1]** to prevent truncation.

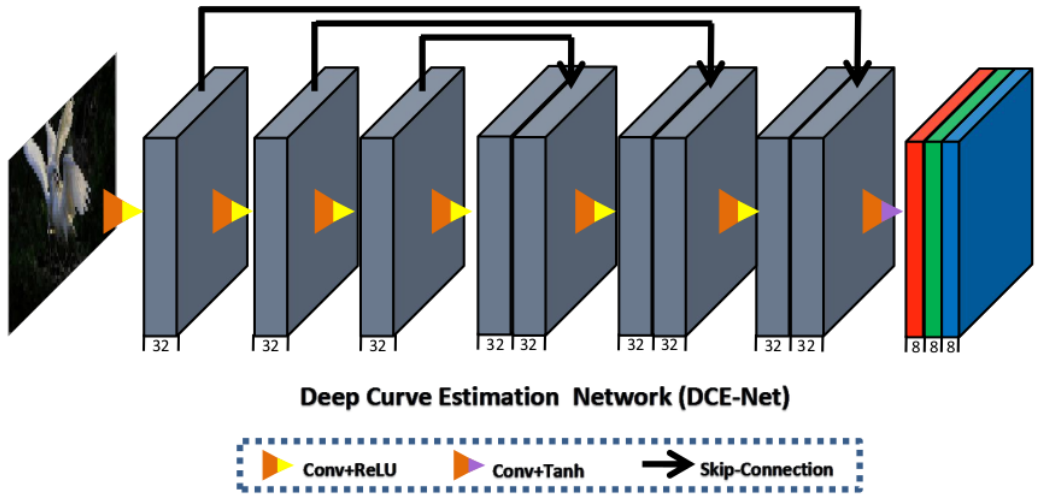
Ensure **monotonicity** to preserve contrast.

Keep the curve **simple & differentiable** for backpropagation.

Applied on RGB channels for better color preservation & reduced over-saturation.

DCE-Net: Deep Curve Estimation Network

- **Purpose:** Learns mapping from low-light images to curve parameter maps.
- **Architecture:**
- **Input:** Low-light image
- **Output:** Pixel-wise curve parameter maps for enhancement
- **Structure:**
  - **7-layer CNN** with symmetrical concatenation
  - **32 convolutional kernels** (3×3, stride 1) per layer
  - **ReLU activation** for intermediate layers
  - **Tanh activation** in the final layer
  - **Output:** 24 parameter maps for **8 iterations** (3 maps per RGB channel).





Applications & Future Research

- Applications: Night vision, medical imaging, security surveillance
- Future directions:
  - Hybrid models (ML + ALSM)
  - Real-time processing improvements
  - Improved color consistency

Conclusion

- **Summary:** Various enhancement techniques improve low-light images using deep learning and curve estimation.
- **ALSM:** A promising method for adaptive light source modeling.
- **Challenges:** Noise reduction, color distortion, and real-time processing remain open issues.

